

GUIDELINES FOR SAFELY USING SOLENOID VALVES

※Please carefully read prior to using the valves.



DANGER

Failure to follow may result in death or serious personal injury.



WARNING

Failure to follow may result in personal injury.



CAUTION

Failure to follow may result in personal injury or machinery equipment damage.

WARNING

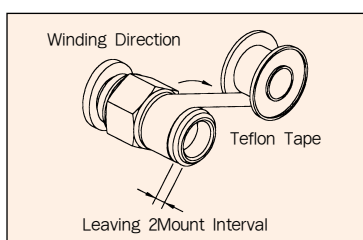
Design, specification, and usage of the compressed air system

- Only the qualified personnel should design, specify, and handle the compressed air system.
- Recommended working condition for each and every product shown in the catalog varies. Implement only the most suitable working condition prior to specifying the system for each product.
- Proper safety features should be installed and secured prior to operating any machinery equipment. The source of disassembling any machinery to avoid any personal injury.

CAUTION

Preparatory factors to consider prior to using the valves

- Verify the accuracy of the specifications of the valves prior to the usage.
- Verify and check that the specifications of the valves are consistent with the requested or ordered specifications.
- Use compressed air to clean any metallic residues, cutting oil, or dust inside of the piping system.
- Distance between the actuator and the valve should be as minimal as possible. The piping distance longer than the minimal leads to inefficiency, waste of compressed air, and slower response time of the valve.
- Use zinc-plated steel tube when steel tubing is required.
- Select and connect the proper ports for accurate piping.
P(1) : Compressed air input connection ports
B(2), A(4) : Actuator connection ports
R2(3), R1(5) : Residual air exhaust ports
- Any and all foreign materials of impurity must be removed at the points of connection or from the inside of valve prior to piping.
- When connecting fittings to the ports, apply Teflon tape to the fitting threads. The first 2 threads of fitting should be free of any Teflon tape. Any fragment or portion of Teflon tape entering the inside of valve will result in the improper operations of the valve.



WARNING

Installation environment for the valves

- Avoid any location where explosive, vibration, and impact factors are present.
- Maintain a sufficient space for a proper ventilation and for any needed repair work when installing valves.
- Use the valves within the allowable range of 5°C ~60°C of the ambient temperature and of the fluid material. In the valves when used under the low temperature working conditions.

CAUTION

Usage conditions for valves

- Installing air filter and oiler prior to piping of valves leads to the extension of the product life.
- Use the purified and dried air only. The source of the compressed air that contains any chemicals, organic raw materials, salt, corrosive materials will render the valves defective and non-operational.
- Excessive air leakage from the compressed air system may reduce the product life of valve.

- Proper filtering system must be installed to prevent any substance of impurity from entering valves. periodical monitoring or replacement of filters is strongly suggested.
- Use turbine oil type or class 2 (ISO VG #32) oil in order to extend the product life, Non-graded oil may be used.
- Use only the recommended oil type. Using non-recommended oil type may reduce the product life of valves due to damage caused by the shrinkage or enlargement of the valve packing. (Do not use spindle oil, machine oil, hydraulic oil, cooking oil, or anti-corrosion oil)
- Do not use the valves in the dusty working condition. If the dusty working condition is unavoidable, place a protective cover over the control unit and place the noise muffler or silencer at the exhaust ports of R2(3) and R1(5).
- Do not allow the valves to be on the stand-by or pause mode for an extended period of time. The stand-by or pause mode of valves should be used briefly and momentarily.
- Solenoid valves are not designed to be suddenly interrupted, stopped, or shut off. Consider and prepare other safety measurements before using the solenoid valves.

WARNING

Electrical factors for the valves

- Allowable range of voltage variation should be within 10% of the recommended voltage for the proper operation of the valves. Any voltage exceeding more than 10% of the recommended voltage will result in the coil damage and the malfunctioning of the valves.
- Turning on or off of Alternate Current (AC) may result in surge current that may have problematic effect to the magnetic circuitry. It is recommended that an alternate plan is implemented to avoid such problematic effect.
- Continuous supply of electricity to the solenoid coil may result in increase of the surface temperature of the coil. In order to avoid the damage to the coil, please verify circuitry design before piping.
- Even if the source of electricity is completely disconnected, the operation of the valves may not come a complete stop due to a residual current in the magnetic circuitry. Reduce the magnitude of the leakage current on both sides by 20% of the rated voltage for AC supply current and by 10% of the rated voltage for DC supply current.

CAUTION

Applicable torque force when piping valves

- Applying excessive torque force will damage the fitting threads when connection to the valves. Use the following guideline of torque force for each applicable port size.

Port size	Applicable Torque N.m(kgf/cm)
M5	2(20)
PT1/8	15(150)
PT1/4	20(200)
PT3/8	25(250)
PT1/2	34(340)
PT3/4	51(510)
PT1	69(690)
PT1 1/4	80(800)
PT1 1/2	85(850)
PT2	120(1200)